Retrieval-Conditional Parsing Score (RCPS): Choosing Document Parsers by Retrieval, Not by Appearance

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Abstract

Practitioners selecting a document parser for retrieval-augmented generation (RAG) typically rank candidates by intrinsic quality metrics—boundary clarity, text-edit fidelity—on the assumption that cleaner parser output retrieves better. On Korean government documents we observe the opposite. In a six-parser, three-retriever evaluation, MoC Boundary Clarity anti-correlates with retrieval (Pearson $r = -0.81$, $n = 5$): the two cleanest-boundary parsers retrieve far below the messier vision–language parsers, and selecting a parser by retrieval rather than by appearance changes Hit@1 by 2.8× (0.197 → 0.549). We make four contributions for production document-RAG. We characterise this parsing–retrieval disconnect in Korean and English and trace its mechanism: under controlled semantic-noise perturbations, boundary clarity stays flat (MinerU, Qwen2.5-VL) or moves non-monotonically (GOT) while retrieval collapses in every family, indicating that intrinsic boundary metrics measure formatting rather than content (C1). A retriever-free coverage diagnostic attributes the gap to a single pipeline layer—20.2% of answers are absent from the parser output, a rate constant across eight chunkers—giving practitioners a rule computable before any retriever is run (C2). We propose RCPS, a retrieval-grounded selection protocol that requires no training and correctly ranks parsers and chunkers that intrinsic metrics misorder; an ablation shows it is not equivalent to single-embedder MRR (C3). Finally, we map the limits of parser-side training: a hidden-state auxiliary loss is sub-threshold and reference-free preference optimisation is negative, while best-of-K fidelity distillation yields a small but reproducible gain (OHR-Bench Hit@5 +1.22 pp), and a matched control shows the retrieval reward itself is unnecessary (C4). We release KoGovDoc-RAG, a Korean document-RAG benchmark, with a reference implementation of RCPS.

1 Introduction

Retrieval-augmented generation (RAG) is increasingly deployed over large collections of real-world documents—government records, enterprise reports, technical manuals—where answering a question means retrieving the right passage from a corpus that exists only as scanned or born-digital PDFs. Every such pipeline begins with a document parser that converts each page to text; that output is then chunked, embedded, and retrieved. Because the parser sits at the head of the pipeline, its errors cascade through every downstream stage, and teams building production systems choose it with care.

The practical question is *how* to choose. Faced with many candidate parsers—OCR engines, layout models, and vision–language models—practitioners rank them by intrinsic parsing-quality metrics that reward clean-looking output: low text-edit distance against a reference, and high boundary clarity (Zhao et al., 2025). The implicit assumption is that cleaner parser output retrieves better.

On Korean government documents, that assumption fails. A practitioner ranking parsers by intrinsic metrics would deploy MinerU (OpenDataLab, 2025), a parser built for high-fidelity OCR whose boundary-clarity score (0.72) is matched only by Marker; yet its retrieval Hit@1 is 0.20, well below the 0.50–0.55 of three lower-scoring vision–language parsers. The pattern is systematic: across six parsers and three retrievers, Boundary Clarity anti-correlates with retrieval at Pearson $r = -0.81$ ($n = 5$). A cross-domain check on the English

OHR-Bench (Zhang et al., 2025) identifies the mechanism—as semantic noise is injected into a parser's output, retrieval degrades sharply while Boundary Clarity barely moves, so the intrinsic metric cannot perceive the content corruption retrieval depends on. Selection by appearance optimises a quantity that is structurally blind to retrievability.

We study how a team should instead select—and, where possible, improve—the parser in a production document-RAG system, and report four contributions.

- **C1 — The parsing–retrieval disconnect and its mechanism.** The parser intrinsic metrics would pick is not the one retrieval wants: parser choice alone swings Hit@1 by 2.8× (0.197 → 0.549), and Boundary Clarity anti-correlates with retrieval (Korean $r = -0.81$, $n = 5$). Controlled noise perturbations on English OHR-Bench show why—boundary metrics track formatting, not content, so noise that destroys retrieval leaves Boundary Clarity flat.

- **C2 — A retriever-free coverage diagnostic.** Holding the parser fixed and varying the chunker, we label each answer covered, split (chunker fault), or absent (parser fault). On Prod, 20.2% are absent against an at-most-2.3% split rate, constant across eight chunkers—a pre-retriever rule: if absent dominates fix the parser, if split dominates fix the chunker.

- **C3 — RCPS, a protocol for selecting parsers and chunkers.** RCPS wraps retrieval MRR in three choices—a held-out Q–A probe, retriever-averaging, and format-invariant relevance—ranking the two parser-side knobs with no training. An ablation shows it is not single-embedder MRR: retriever-averaging corrects the top ranking a single embedder gets wrong.

- **C4 — A bounded map of parser-side training.** Two natural formulations fail (hidden-state auxiliary loss sub-threshold, reference-free preference optimisation negative). Discrete-output preference training gives a small cross-domain gain (OHR Hit@5 +0.85 pp), but a matched best-of-K control ranked by edit-distance to ground truth reproduces it (+1.22 pp)—the lever is text-fidelity distillation, not the retrieval reward.

We release **KoGovDoc-RAG**, a Korean document-RAG benchmark of 663 question–answer pairs over 294 pages, together with a reference implementation of RCPS, the RADP-Distill checkpoint (the recommended parser-side lever), and the RADP-aux / RADP-DPO checkpoints used for the negative and controlled comparisons.

2 Related Work

Diagnosing the parsing–retrieval gap. Prior work documents that parser quality affects downstream retrieval but does not close the gap on the parser side. OHR-Bench (Zhang et al., 2025) evaluates the cascading impact of OCR noise on RAG; EnterpriseDocBench (EnterpriseDocBench authors, 2026) reports a weak parsing-quality–retrieval correlation; and concurrent analyses (Anonymous, 2026) reach similar conclusions. None proposes a training-time intervention on the parser itself.

Training-time methods, by pipeline layer. Existing parser-adjacent training operates at every layer except the parser. Chunking methods decide boundaries after parsing (Günther et al., 2024; Duarte et al., 2024; Zhao et al., 2024, 2025); embedding-side methods train the retriever contrastively (Conti et al., 2025; LMAR authors, 2025); and retrieval- or reader-side methods tune the retriever or generator (Nguyen et al., 2024; Chia et al., 2025; Yan et al., 2025). To our knowledge no prior work trains the parser itself on a retrieval signal—the gap our C4 occupies—while our primary contributions (C1–C3) sit upstream of any training.

3 Method

3.1 RCPS: Retrieval-Conditional Parsing Score

RCPS ranks parsers by what downstream retrieval does with their output rather than by how clean that output looks. It is not a new similarity function but an evaluation protocol that wraps ordinary retrieval MRR in three choices: (i) it is *extrinsic*, scoring on the parsed corpus together with a held-out Q–A probe rather than on text alone; (ii) it is *retriever-agnostic*, averaging over several embedders so the ranking does not hinge on which one sits in the production stack; and (iii) it uses *format-invariant relevance*, counting a chunk as relevant if and only if its text contains the answer span, however the parser formatted it. The contribution is the protocol—what to measure, on what probe, and how to judge relevance—not the scoring function; this is also why RCPS is not single-embedder MRR, which yields a different and less stable parser ranking (§4.3).

Given a parser P, a Q–A set $D = \{(q_i, a_i, \text{page}_i)\}$, a set of retrievers R, and cutoffs K, RCPS averages MRR across the cross-product:

$$\text{RCPS}(P, D, R, K) = \frac{1}{|R||K|} \sum_{r \in R} \sum_{k \in K} \text{MRR@}k\big(r, \text{chunks}_P(D), \{q_i\}\big). \quad (1)$$

We use $R = \{$BGE-M3 (Chen et al., 2024), multilingual-e5-large (Wang et al., 2024), Qwen3-Embedding-8B (Qwen Team, Alibaba, 2025a)$\}$ and $K = \{1, 5, 10\}$. A chunk is relevant for a query if and only if its source page matches the answer's source page and the gold answer span is a substring of the chunk under whitespace- and markdown-insensitive normalisation. The retriever average makes the ranking robust to embedder choice: a parser that wins on one retriever but loses on another does not dominate. In practice a team runs RCPS on a few hundred held-out Q–A, with no training, to choose a parser or chunker that intrinsic metrics rank incorrectly; Figure 2 summarises the protocol together with the failure each of its three choices removes. The implementation is released.

3.2 Coverage Diagnostic — Locating the Gap (Parser vs Chunker)

RCPS scores the parser, chunker, and retriever jointly, so a low score does not localise the fault. We separate the parser from the chunker with a retriever-free diagnostic: holding the parser output fixed and varying the chunker, we classify each gold answer as *covered* (inside a single chunk, retrievable); *split* (present in the page text but cut across chunks—a chunker fault, recoverable with overlap); or *absent* (not in the parser output at all—a parser fault, unrecoverable by chunking). Coverage is the ceiling on retrieval, since split and absent answers score zero; the split/absent breakdown attributes the gap to a pipeline layer and decides whether a parser-side fix can help (§4.2). Relevance reuses RCPS's format-invariant matching.

3.3 Parser-Side Training: Auxiliary Loss and Discrete-Output DPO

When the diagnostic points to the parser (§4.2), we train it on a retrieval signal in two natural ways. **(a) Hidden-state auxiliary loss (RADP-aux):** we add to the parsing loss an InfoNCE term aligning the parser's pooled answer-span hidden state with the frozen BGE-M3 embedding of the gold chunk ($\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{parse}} + \lambda \mathcal{L}_{\text{contrast}}$); this differentiable surrogate reaches the deployed markdown only through $\mathcal{L}_{\text{parse}}$ and is subthreshold (§4.4). **(b) Discrete-output DPO (RADP-DPO):** we optimise the markdown directly (Rafailov et al., 2023)—sampling K parses per page, scoring each by a page-local RCPS, forming preference pairs (gap ≥ 5 pp), and training with a LoRA-toggle reference (Hu et al., 2022) (π_θ adapter-on, π_{ref} adapter-off), the reward sharpened over three milestones R1→R2→R3 (Appendix A); reference-free SimPO (Meng et al., 2024) is a control. **(c) Reward-agnostic control (RADP-Distill):** the identical best-of-K pipeline but ranking candidates by edit-distance to the ground-truth markdown instead of RCPS; if it matches RADP-DPO, the retrieval reward is unnecessary (§4.4).

4 Experiments

4.1 Setup

We construct **KoGovDoc-RAG**: 663 Q–A pairs over 294 pages of Korean government documents, generated with the OpenAI `gpt-5.4-2026-03-05` model and verified by an LLM-as-judge against the human-curated ground-truth markdown (100 stratified eval Q–A sampled for verification, 94/100 accept). For RADP-aux's full-scale training we additionally generate 6,164 Q–A on the 2,667-page Prod training set. Cross-domain replication uses OHR-Bench (Zhang et al., 2025) (seven domains, 2,264 verbatim-answerable Q–A; its Law + Manual subset of 1,043 Q–A is used only for the data-mix-sensitivity comparison in §4.2) across the three released parser outputs (gt, MinerU, Qwen2.5-VL) plus twelve controlled noise perturbations.

Throughout, **Prod** denotes our production parser, a Qwen3-VL-2B model (Qwen Team, Alibaba, 2025b) fine-tuned for Korean document parsing; all training deltas are measured against it. RADP-aux uses a 73-page held-out fold (202 Q–A); RADP-DPO, SimPO, and the §4.4 analysis use the combined 242-page fold (train ∪ eval, 663 Q–A), with preference pairs built only from the 169-page train fold. The same 663 Q–A occupy 242 of the 294 KoGov pages; the coverage diagnostic (§4.2) runs over Prod's full 294-page output, the extra 52 Q–A-free pages acting as distractors. Parses for the 12-variant comparison (§4.4–§4.4) use deterministic decoding (temperature 0, max 1536 tokens). We fine-tune with LoRA ($r = 8$, $\alpha = 32$), sweeping $\lambda \in \{0, 0.1, 0.3, 0.5\}$ for RADP-aux ($\lambda = 0$ drops the contrastive term; the aux recipe itself shifts output from Prod, Table 5, so the sweep is read within the aux family). RCPS uses the three retrievers and cutoffs of §3.1; we report paired percentile-bootstrap 95% CIs (1,000 resamples), preserving Q–A pairing across systems.

4.2 The Parsing–Retrieval Disconnect and Coverage Diagnostic (C1–C2)

Korean government documents (Figure 1; full grid in Appendix D). RCPS spans 0.07–0.58 across the six parsers: the vision–language parsers cluster at the top (0.53–0.58) and the OCR systems trail (0.07–0.21). Within that top cluster the 30B teacher ranks first (RCPS 0.584), with the deployable 2B Prod a negligible 0.001 behind (0.583); Figure 1b therefore contrasts Prod, the model a team would ship, against the appearance-ranked MinerU. The intrinsic Boundary Clarity metric (Zhao et al., 2025) anti-correlates with RCPS at Pearson $r = -0.81$ ($n = 5$, excluding PaddleOCR, whose Boundary Clarity is undefined because MoC detects zero boundaries in its output). The two cleanest-boundary parsers, MinerU and Marker (BC ≈ 0.72), land near the bottom of the retrieval ranking (RCPS 0.07–0.21), while the lower-BC vision–language parsers (BC 0.52–0.62) retrieve best (0.53–0.58). Marker is evaluated on a 38-page subset rather than the full corpus; dropping it leaves the anti-correlation unchanged in sign and magnitude ($n = 4$, $r = -0.74$). MoC's companion metric, Chunk Stickiness, is similarly disconnected ($r = +0.26$, $n = 5$; with CS oriented so that lower is more cohesive, this is the same direction as BC—more cohesive intrinsic structure tracks worse retrieval). Neither intrinsic axis predicts RCPS.

A single-domain anti-correlation could be a quirk of one language or document type, so we test cross-domain.

Cross-domain: the mechanism. Across all seven OHR-Bench domains (2,264 Q–A) we evaluate 15 parser-output variants: three released parser outputs (gt, MinerU, Qwen2.5-VL), three formatting-noise perturbations, and nine semantic-noise perturbations (GOT, MinerU, Qwen2.5-VL $\times$ mild, moderate, severe). Within each semantic-noise family, Boundary Clarity fails to track noise severity while RCPS collapses (per-variant values in Appendix E). For the MinerU family, BC stays in 0.71–0.74 from clean to severe while RCPS falls from 0.50 to 0.24 (−51%); GOT shows the same pattern (RCPS 0.38 $\rightarrow$ 0.26) and Qwen2.5-VL is more noise-robust (0.47 $\rightarrow$ 0.43, −8%). The per-variant BC and RCPS values are in Appendix E. Semantic noise that destroys retrievable content does not lower Boundary Clarity: the intrinsic metric sees formatting, not content.

The aggregate cross-variant correlation is sensitive to the document mix: Pearson BC↔RCPS across all 15 variants is −0.35 on Law + Manual alone but +0.25 on the full seven-domain corpus. We therefore report the per-family mechanism, which reproduces in every domain, as the robust finding, and treat the cross-variant scalar—which conflates parser families of differing noise-robustness—as illustrative only.

Locating the gap: parser or chunker? (Table 1). The disconnect shows that intrinsic metrics mislead, but not which pipeline

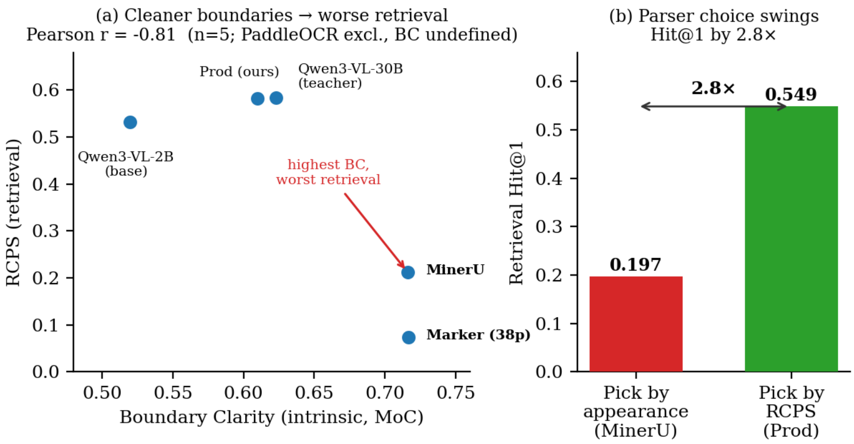

Figure 1: The parsing–retrieval disconnect on Korean government documents. (a) Boundary Clarity (intrinsic, MoC) anti-correlates with RCPS across parsers (Pearson $r = -0.81$, $n = 5$; PaddleOCR excluded as its BC is undefined): the cleanest-boundary parsers MinerU and Marker land near the bottom, while the lower-BC vision–language parsers retrieve best. (b) Choosing the parser by RCPS (Prod) rather than by appearance (MinerU) changes retrieval Hit@1 by 2.8× (0.197 → 0.549).

Chunker	covered	split	absent
md_h3	79.8%	0.0%	20.2%
parser_native	78.1%	1.7%	20.2%
fixed500	77.5%	2.3%	20.2%
fixed500_ov200	79.8%	0.0%	20.2%

Table 1: Answer-coverage diagnostic on Prod's output (294 pages, 663 KoGov Q–A; text matching only, no retriever). The absent rate (a parser fault) is constant at 20.2% across all eight chunkers (four representative rows shown), the required boundary-independence check; the split rate (a chunker fault) is at most 2.3% and vanishes under overlap. Chunkers: md_h{1,2,3} = markdown-header splitting at heading depth k; parser_native = the parser's own segmentation; fixed N[_ov$M$] = fixed N-character windows, optionally with M-character overlap.

layer is at fault. Applying the §3.2 classification with no retriever, on Prod's output (294 pages, 663 Q–A) 20.2% of answers are absent and at most 2.3% are split, and the absent rate is constant across all eight chunkers we test—exactly the boundary-independence a parser fault must show. The gap is therefore a parser problem: roughly one answer in five is never produced, so no re-chunking can recover it, and a parser-side intervention is the correct lever. This both motivates the training in §4.4 and yields the practitioner rule: run this diagnostic first, and if absent dominates, fix the parser; if split dominates, fix the chunker.

4.3 RCPS Selects Both Parsers and Chunkers (C3)

A selection protocol must separate the alternatives a practitioner actually compares, across both knobs they control. For **parsers**, the six-parser grid of §4.2 (Figure 1, Appendix D) is itself an RCPS ranking: it tells a team that Prod (RCPS 0.583, Hit@1 0.55) beats MinerU (0.21, 0.20) by 2.8×, the ordering Boundary Clarity inverts. For **chunkers**, on a fixed Prod output RCPS separates four strategies cleanly—markdown-header (md-h3, RCPS 0.593) > parser-native (0.583) > LumberChunker (Duarte et al., 2024) (0.557) > fixed-size (0.535)—whereas intrinsic boundary metrics rank them inconsistently, or rank fixed-size highest because its boundaries are clean by construction. Because RCPS is retriever-averaged and format-invariant, one probe set of a few hundred Q–A ranks both the parser choice and the chunker choice a team makes independently.

RCPS is not single-embedder MRR (Table 2). Stripping the three protocol choices from the six-parser KoGov grid shows they are not cosmetic. Dropping retriever-averaging—scoring on a single embedder (BGE-M3)—inverts the top parser choice: naive single-embedder MRR ranks Prod first, while full RCPS ranks the Qwen3-VL-30B teacher first (Table 2). Format-invariant matching, by con-

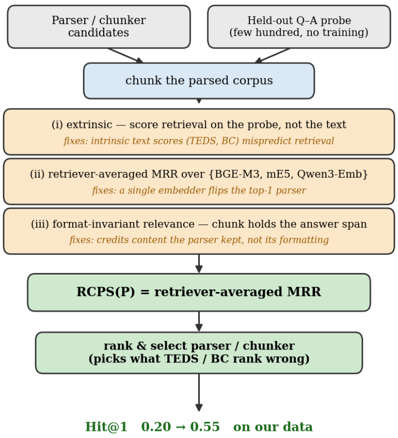

Figure 2: RCPS as a protocol, not a new scoring function. It wraps ordinary retrieval MRR in three choices, each removing a failure mode of intrinsic metrics: (i) it scores retrieval on a held-out Q–A probe rather than on the text, since text scores (TEDS, BC) mispredict retrieval; (ii) it averages over retrievers, since a single embedder can flip the top-ranked parser (Table 2); and (iii) it uses format-invariant relevance, crediting the content a parser keeps rather than its formatting. Run with no training on a probe of a few hundred Q–A, it ranks the parser and chunker a team would deploy.

Protocol configuration	Top pair	τ vs RCPS
naive MRR (1 embedder, fmt-sensitive)	Prod $\succ$ 30B	0.87
+ retriever-averaging	30B $\succ$ Prod	1.00
+ format-invariant only	Prod $\succ$ 30B	0.87
full RCPS (3 retrievers, fmt-invariant)	30B $\succ$ Prod	—

Table 2: RCPS protocol ablation on the KoGov six-parser grid (only the inverted top pair is shown; the lower ranks Qwen3-VL-2B $\succ$ MinerU $\succ$ PaddleOCR $\succ$ Marker are identical across all configurations). Retriever-averaging is the single choice that flips the top pair (rows 2,4 vs 1,3); format-invariance shifts scores but not order. Kendall τ is over the full six-parser ranking; "30B" is the Qwen3-VL-30B teacher.

trast, shifts every parser's score by about 0.02–0.03 but does not reorder the grid. The inversion is between two near-tied parsers (RCPS 0.584 versus 0.583), so the operational claim is precise: single-embedder MRR cannot reliably resolve the top parser a team would deploy—exactly the decision RCPS is run to make. The ordering disagreement (Kendall $\tau = 0.87$) substantiates the claim that RCPS is a protocol, not a relabelled MRR.

4.4 Parser-Side Training Is a Bounded, Reward-Agnostic Lever (C4)

With one answer in five absent from the parser (§4.2), a parser-side fix is the right lever. Of the two natural directions, only discrete-output preference training (RADP-DPO, §3.3) produces a deployable effect—the hidden-state auxiliary loss is sub-threshold and reference-free SimPO is uniformly negative (Appendix B). On KoGov, RADP-DPO improves Hit@5 by +1.96 to +2.11 pp, but at $n = 663$ every two-sided interval spans zero, so we treat KoGov as exploratory and rely on the pre-specified OHR-Bench replication, where training signal, metric, and language are disjoint. There the representative variant R2 gives Hit@5 +0.85 pp $[+0.35, +1.43]$, two-sided significant across all seven domains (Table 4).

The retrieval reward is unnecessary. A matched RADP-Distill control—identical but ranking pairs by edit-distance to ground truth rather than RCPS—reproduces the gain (OHR Hit@5 +1.22 pp $[+0.35, +2.15]$; KoGov +2.61 pp), indistinguishable from RADP-DPO at the powered endpoint. The lever is therefore fidelity distillation, not the retrieval reward—a cheaper route to the same target.

The mechanism is text fidelity, not chunk boundaries. Across the 12 systems (Appendix C, Table 5), every DPO variant cuts TextNED from Prod's 0.240 to 0.16–0.19 and RADP-Distill cuts it furthest (0.158), while the MoC chunking signature (BC 0.60–0.66, CS $\approx$ 0.47) does not move; RADP-aux, which instead *raises* TextNED, stays sub-threshold. The gain comes from *what* is parsed, not *how* it is chunked—§4.2's intrinsic-metric blindness seen from the training side—and replicates cross-domain (English R2 TextNED −1.36%, significant). RADP-DPO is thus best read as best-of-K self-distillation: training amortises into the weights the higher-fidelity parses the model could already occasionally produce.

5 Discussion and Conclusion

Selection, not training, is the headline. The largest and most certain effect here is not a training method: choosing a parser by retrieval (RCPS) rather than by intrinsic metrics changes Hit@1 by 2.8× (§4.2). A team that adopts only the RCPS protocol already avoids shipping the parser that looks best yet retrieves near the bottom. Parser-side training is a secondary, bounded lever of about 1 pp Hit@5, and the retrieval reward buys nothing over plain fidelity distillation (§4.4).

Why the diagnosis and the mechanism agree. C1 finds the cleanest-boundary parser retrieves among the worst; §4.4 finds training helps by producing text closer to ground truth. Both follow from splitting "clean" into two axes—Boundary Clarity measures formatting, TextNED measures content fidelity (whether the text holds the answer span). MinerU is formatting-clean but loses content; training improves fidelity while leaving the formatting signature unchanged, which is exactly why intrinsic boundary metrics mispredict retrieval. This also refines our starting hypothesis: we expected training to move chunk boundaries, but what moves is text fidelity.

Playbook and conclusion. For practitioners: (1) evaluate parsers with RCPS, not intrinsic metrics—a few hundred held-out Q–A, no training, reorder parsers as the retriever sees them (on our data a 0.20 → 0.55 Hit@1 decision); (2) if you train, distil the discrete output toward clean ground-truth text—the auxiliary loss and reference-free SimPO are sub-threshold, and a retrieval reward is unnecessary; (3) the gain concentrates on factoid queries and is neutral on tabular ones, so layout-heavy query mixes should instead look to the chunker or embedder. We documented the parsing–retrieval disconnect in two languages, proposed RCPS for selecting parsers and chunkers by retrieval rather than appearance, and bounded what parser-side training adds. Code, data, and checkpoints are released.

Limitations

Single primary language and statistical power. The C1 diagnostic is strongest in Korean ($n = 5$, $r = -0.81$). On English OHR-Bench the aggregate cross-variant scalar is sign-sensitive to document mix ($r = -0.35$ on Law + Manual, +0.25 on the full seven-domain corpus), so directional consistency holds only for the per-family noise mechanism (§4.2), which replicates in every domain—not for the scalar correlation. The English evidence is also built on three real parser outputs plus twelve controlled perturbations rather than fifteen independent parsers. With these sample sizes the correlations are illustrative rather than inferential.

Q–A generation and construct validity. The KoGov Q–A were produced by a large language model and verified by an LLM-as-judge against the human-curated ground-truth markdown (94/100 accept on a 100-Q–A stratified sample; full human verification at train scale was cost-prohibitive). Synthetic queries bound what RCPS can claim, but the exposure is narrow: our comparative findings—the 2.8× parser-choice swing and every parser/chunker ranking—hold the Q–A set fixed across systems and so are internally valid regardless of how queries were generated, and the confirmatory cross-domain test (§4.4) uses OHR-Bench's own externally curated Q–A. We release the frozen KoGov eval set for audit.

KoGov is exploratory; OHR-Bench is confirmatory. The headline KoGov cell (R2 +1.96 pp, Table 3) has a paired 95% interval of $[-1.06, +5.03]$—two-sided non-significant and selected from a multi-cell scan—so we claim no two-sided significance on KoGov. The confirmatory evidence is the prespecified OHR-Bench replication (R2 +0.85 pp $[+0.35, +1.43]$, $n = 2{,}264$). A larger KoGov fold ($\geq 1{,}500$ Q–A) would be needed for a powered two-sided result.

Candidate pool and untested paradigms. Preference pairs are sampled from Prod itself; an alternative pool (e.g. chosen/rejected pairs from other parsers) might shift the mechanism away from GT-fidelity, which we did not test. We also test only the hidden-state auxiliary loss, discrete-output DPO with a reference policy, and reference-free SimPO; token-level RL, multi-task interleaving, curriculum schedules, and chunk-granularity oracle distillation are untested. Finally, RADP-DPO is a parser-layer intervention; chunker-, embedder-

, and reranker-side training on RCPS-graded chunks are complementary directions left to future work.

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A RADP-DPO Milestone Construction

For each train page we sample K alternative parses from the production parser, chunk and score each by a page-local RCPS (the page's questions against the parse's own chunks plus distractors), and form preference pairs (chosen = higher-RCPS parse, gap ≥ 5 pp). The reward is sharpened across three milestones: **R1** ($K = 2$ parses, uniform distractors, $\beta = 0.1$) $\rightarrow$ **R2** (a warm-started iterative round, $\beta = 0.05$) $\rightarrow$ **R3** ($K = 14$ parses with a full-corpus hard-negative pool—the other-page chunks closest to the page's queries—widening the candidate-score gap from about 0.05 to 0.56). We train with a LoRA-toggle reference (π_θ = LoRA on, π_{ref} = LoRA off):

$$\mathcal{L}_{\text{DPO}} = -\log \sigma\Big(\beta\big[(\log \pi_\theta(c) - \log \pi_\theta(r)) - (\log \pi_{\text{ref}}(c) - \log \pi_{\text{ref}} \quad (2)$$

The reference-free SimPO control ($\beta = 2.0$, $\gamma = 1.0$) removes the reference policy entirely.

Variant (vs Prod = 0.6863)	Hit@5	ΔHit@5 [95% CI]	$P[\Delta>0]$	ΔRCPS
RADP-DPO-v5 (R3)	0.7074	+2.11 [−0.96, +5.13]	0.913	+1.72
RADP-DPO-v1 (R1)	0.7069	+2.06 [−0.96, +5.13]	0.907	+0.57
RADP-DPO-v4 (R2)	0.7059	+1.96 [−1.06, +5.03]	0.897	+0.47
RADP-SimPO (ctrl)	0.6793	−0.70 [−3.77, +2.31]	0.321	−1.56

Table 3: RADP-DPO milestone progression and the SimPO control on parser-native chunking, 242-page / 663-Q–A fold, 10k paired percentile bootstrap, all against the same Prod baseline (Hit@5 = 0.6863). Macro Hit@k averages the three retrievers of §3.1. All three milestones are strong directional positives ($P[\Delta>0] \geq 0.90$) but two-sided non-significant at this fold size; RADP-SimPO is uniformly negative. The RADP-Distill headline (KoGov ΔHit@5 +2.61 pp) is in §4.4 and Table 4. Across three Prod seeds the standard deviation of the mean ΔHit@5 is 0.90 pp.

Δ vs Prod (pp)	Hit@1	Hit@5	Hit@10	MRR@10	nDCG@5
RADP-Distill	**+0.88**	**+1.22**	**+1.32**	**+1.01**	**+1.05**
RADP-DPO-v4 (R2)	+0.53	+0.85	+0.81	+0.70	+0.74
RADP-DPO-v5 (R3)	+1.31	+1.03	+0.81	+1.17	+1.15

Table 4: OHR-Bench cross-domain Δ vs Prod, in pp (2,264 Q–A over seven English domains; three-retriever macro; 1,000-resample paired bootstrap). The Hit@5 deltas are two-sided significant (95% CIs: RADP-Distill [+0.35, +2.15], R2 [+0.35, +1.43], R3 [+0.24, +1.84]). The other columns are monotone re-cuts of the same per-system ranked list, reported only because practitioners use different cutoffs. RADP-Distill leads at the primary metric Hit@5; R3 reaches higher Hit@1 and MRR@10 point estimates but with overlapping CIs, so the retrieval reward is unnecessary at the powered endpoint (§4.4).

B KoGov DPO Progression, RADP-aux Sweep, SimPO, and Robustness

On the 242-page KoGov fold (10,000-resample paired bootstrap), the three RADP-DPO milestones improve Hit@5 on parser-native chunking over Prod along the reward-sharpening axis: R1 +2.06 pp [−0.96, +5.13], R2 +1.96 pp [−1.06, +5.03], R3 +2.11 pp [−0.96, +5.13] (all $P[\Delta>0] \approx 0.90$). At $n = 663$ every two-sided interval spans zero, so these are exploratory: the Hit@5 / parser-native cell was selected from a multi-cell scan over (chunker × retriever × k) with no family-wise correction. The reward-agnostic RADP-Distill control, evaluated on the same fold, reaches +2.61 pp and is the headline; its powered confirmation is the OHR-Bench result (Table 4, +1.22 pp).

RADP-aux is sub-threshold. The auxiliary loss peaks at $\lambda = 0.1$ (+1–2 pp RCPS) then declines monotonically as the contrastive objective competes with $\mathcal{L}_{\text{parse}}$ over the same LoRA parameters; every Δ-versus-control interval includes zero and the one-sided $P[\Delta>0]$ never approaches 0.90. The hidden-state route does not reach the deployed markdown—only the discrete output does.

SimPO is negative. The reference-free length-normalised variant gives uniformly negative deltas (−0.7 to −1.7 pp Hit@5), indicating the DPO reference policy is load-bearing: without it the parser drifts from the production distribution before any preference signal can compound.

Robustness. The OHR-Bench gain survives two checks. It is equal or larger on the two retrievers held out from preference scoring (ml-e5-large +2.4 pp, Qwen3-Emb +2.3 pp, versus the BGE-M3 scorer +1.5 pp), ruling out a BGE-overfit artefact; and it concentrates on factoid queries (+3.1 pp) while neutral-to-slightly-negative on tabular ones, consistent with the text-fidelity mechanism.

C Chunk-Level Mechanism Statistics

We measure four chunk-level statistics on the 242-page fold for all 12 systems plus the R3 variant (Table 5): MoC Boundary Clarity (BC, adjacent-chunk discontinuity), MoC Chunk Stickiness (CS, within-chunk cohesion), normalised edit distance to the ground-truth markdown (TextNED, character-level Levenshtein distance ÷ longer-string length), and chunking shape.

The signal. All five RADP-DPO variants reduce TextNED from Prod's 0.240 to 0.163–0.185, a 23–32% move toward the ground-truth markdown; the two replicating positive variants (DPO-v1, DPO-v4) achieve the largest reductions (0.167, 0.163). RADP-aux instead increases TextNED (0.318–0.626) because its hidden-state objective degrades surface text. TextNED is positively associated with Hit@5 (R3 the one DPO-cluster exception), and RADP-Distill reduces it furthest

Variant	len	c/pg	clen	BC↑	CS↓	TextNED↓
Prod (ref)	1385	4.82	274	0.630	0.474	0.240
RADP-Distill	1597	5.29	291	0.641	—	**0.158**
aux λ=0.0	799	2.96	245	0.553	0.473	0.626
aux λ=0.1	1380	4.05	321	0.652	0.484	0.423
aux λ=0.3	1930	6.73	272	0.470	0.475	0.332
aux λ=0.5	2035	5.82	327	0.556	0.470	0.318
DPO-v1	1606	4.97	311	0.646	0.474	**0.167**
DPO-v2	1594	5.06	304	0.648	0.475	0.168
DPO-v3	1689	5.27	304	0.601	0.469	0.182
DPO-v4 (R2)	1610	4.83	321	0.647	0.476	**0.163**
DPO-v5 (R3)	1514	4.88	300	0.656	0.485	0.185
SimPO	1703	5.31	305	0.601	0.470	0.186
DPO-v1-s123	1633	4.92	320	0.655	0.476	0.171
DPO-v1-s999	1620	4.69	332	0.654	0.478	0.174

Table 5: Chunk-level mechanism statistics on the 242-page fold (len = parse length, c/pg = chunks per page, clen = chunk length). RADP-Distill attains the lowest TextNED of all (0.158 < R2's 0.163 < Prod's 0.240) with BC unchanged (0.641, within Prod's range), confirming that text fidelity, not chunk structure, is the operative axis (CS was not measured for RADP-Distill, which uses a fixed chunking scheme). Among retrieval-reward variants, R2 has the lowest TextNED (0.163); R3's KoGov TextNED (0.185) does not beat it.

(0.158) while leading on retrieval—the cleanest demonstration that text fidelity, not the retrieval reward, is the lever.

Cross-domain replication. On all 4,040 English OHR-Bench pages, zero-shot Prod reaches TextNED 0.192—below its 0.240 on Korean, so English leaves less fidelity headroom. R2 reduces it to 0.1894 (a -0.0026 delta, -1.36%, 95% CI $[-0.0043, -0.0010]$, two-sided significant). R3 pushes the English value marginally lower (0.184) yet does not improve KoGov fidelity, which is why R2 is the headline variant.

The non-signal. The MoC chunking signature is unchanged on both axes: BC for all DPO variants lands in 0.60–0.66 (Prod 0.63) and CS in 0.469–0.485 across every DPO and SimPO variant (Prod 0.474; CS not measured for RADP-Distill); chunks-per-page (4.69–5.27) sits within Prod's range, and chunk length (300–332) runs 10–21% above Prod's 274 while neither MoC axis (BC, CS) moves. Training does not produce a novel chunking signature; the gain comes from what is parsed, not how chunks are split.

Secondary patterns. The gain concentrates on factoid queries—where the verbatim answer span must land inside a chunk—and is neutral on tabular ones. It is larger on the held-out retrievers than on the BGE-M3 scorer, because Prod's text was already BGE-aligned, confirming a parser-level effect rather than a BGE-overfit. RADP-aux stays sub-threshold because its hidden-state signal reaches the deployed markdown only through diffuse $\mathcal{L}_{\text{parse}}$ back-flow, never localising to the surface tokens the retriever rewards.

D KoGov Parser Grid

Parser	BC	CS	RCPS	Hit@1
Qwen3-VL-30B (teacher)	0.623	3.38	**0.584**	0.545
Prod (ours, 2B)	0.610	3.07	0.583	0.549
Qwen3-VL-2B (base)	0.520	3.74	0.532	0.500
MinerU	0.716	2.81	0.212	0.197
PaddleOCR	—	3.46	0.140	0.125
Marker (38p)	**0.717**	3.41	0.073	0.068

Table 6: The §4.2 disconnect grid, visualised in Figure 1: KoGov, BC versus RCPS, Pearson $r = -0.81$ ($n = 5$, excluding PaddleOCR, whose Boundary Clarity is undefined). Marker is evaluated on a 38-page subset; dropping it gives $n = 4$, $r = -0.74$, same direction. Chunk Stickiness (CS, MoC; lower = more cohesive) is similarly uninformative ($r = +0.26$, $n = 5$). This parser-grid CS is on a different scale from the per-variant mechanism CS in Table 5 and is not directly comparable. RCPS is averaged over three retrievers.

E OHR-Bench Noise-Family Grid

Family	Severity	BC	RCPS
MinerU	clean	0.735	0.496
MinerU	mild	0.708	0.413
MinerU	moderate	0.715	0.351
MinerU	severe	0.712	0.243
Qwen2.5-VL	clean	0.619	0.467
Qwen2.5-VL	mild	0.614	0.462
Qwen2.5-VL	moderate	0.616	0.460
Qwen2.5-VL	severe	0.610	0.429
GOT	mild	0.634	0.383
GOT	moderate	0.495	0.338
GOT	severe	0.650	0.257

Table 7: Per-family Boundary Clarity and RCPS across semantic-noise severity on the seven OHR-Bench domains (the source for the §4.2 noise numbers). Within each family BC stays roughly flat while RCPS falls: MinerU 0.496 $\rightarrow$ 0.243 (-51%), GOT 0.383 $\rightarrow$ 0.257, Qwen2.5-VL 0.467 $\rightarrow$ 0.429 (-8%). GOT has no clean baseline among the released outputs. The table lists the three semantic-noise families (11 of the 15 evaluated variants); the ground-truth output and the three formatting-noise perturbations, which enter only the aggregate cross-variant scalar (§4.2), are in the released grid.